

Nanditha Thupakula

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PROFESSIONAL SUMMARY

AI Engineer specializing in **Generative AI, Large Language Models (LLMs), RAG, Agentic AI, and machine learning systems**. Experienced in developing LLM-powered applications, retrieval pipelines, model evaluation workflows, REST APIs, and production-oriented AI solutions. Proficient in **Python, PyTorch, Hugging Face, LangChain, LangGraph, FastAPI, AWS, Docker, and vector databases**, with hands-on experience in prompt engineering, fine-tuning, embeddings, inference optimization, and AI system deployment.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++, JavaScript, Bash

AI/ML: Generative AI, LLMs, Transformers, Fine-tuning, PEFT, Quantization, Prompt Engineering, Function Calling, Agentic AI, RAG, NLP, Semantic Search, Embeddings, Tokenization, Model Evaluation, Inference Optimization, LLMOps

Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, CrewAI, LlamaIndex, vLLM, ONNX, FastAPI, Streamlit, Scikit-Learn

Cloud/MLOps: AWS (SageMaker, S3, Lambda), Docker, Kubernetes, Terraform, MLflow, CI/CD, GitHub Actions, LangSmith, Prometheus, Grafana

Databases/Data: Pinecone, ChromaDB, FAISS, Weaviate, PostgreSQL, MongoDB, Kafka, DVC, Pandas, NumPy, Power BI, ETL

Evaluation: RAGAS, BLEU, ROUGE, Retrieval Evaluation, Model Benchmarking, Prompt Evaluation

EDUCATION

University of Cincinnati

Master of Engineering in Artificial Intelligence

Cincinnati, Ohio

Aug 2024 – Apr 2026

Madanapalle Institute of Technology and Science

Bachelor of Technology in Electronics and Communication Engineering

India

2019 – 2023

EXPERIENCE

University of Cincinnati

Event Staff – Technical Support

Cincinnati, Ohio

Dec 2024 – Apr 2026

- Diagnosed and resolved real-time faults across AV, networked, and embedded event systems using structured root-cause analysis, reducing downtime during high-traffic university events.
- Provided rapid technical troubleshooting and incident response under time-critical conditions, strengthening systems reliability, operational monitoring, and structured problem-solving skills.

MindTree

Software Engineer

Bengaluru, India

Nov 2023 – Jul 2024

- Developed cloud-native microservices on AWS using Terraform infrastructure-as-code, supporting distributed systems with 99.9% availability and automated monitoring and alerting workflows.
- Fine-tuned transformer-based models for text classification and optimized inference pipelines through latency benchmarking and model evaluation, improving inference speed by 10%.
- Built Python-based data preprocessing and model evaluation workflows for machine learning applications, improving reproducibility and deployment readiness.

MindTree

Java Developer Trainee

Bengaluru, India

Jun 2023 – Nov 2023

- Engineered REST API services using Spring Boot for high-concurrency applications and integrated automated analytics workflows into CI/CD pipelines, improving processing efficiency by 15%.

PROJECTS

Structura AI – Civil Document Assistant

Python, LangChain, ChromaDB, Ollama, Qwen3, Streamlit

- Built a domain-adapted RAG assistant for querying inspection reports, design manuals, specifications, and building codes, generating context-grounded answers with source-page citations.
- Implemented document ingestion, text chunking, embedding-based retrieval, prompt orchestration, and local LLM inference using Ollama and Qwen3 for domain-specific question answering.

Autonomous Multi-Agent Researcher

CrewAI, LangGraph, GPT-4o, FastAPI, vLLM

- Architected a multi-agent research system using LangGraph workflows, function calling, RAG, and error-correction loops, reducing hallucinations by 30% through iterative validation and evaluation.
- Implemented FastAPI services and model-serving workflows with vLLM while applying RAGAS-based evaluation to monitor answer quality and retrieval performance.

Enterprise Knowledge Assistant

LangChain, Pinecone, FastAPI, LangSmith, ONNX

- Built a production-oriented RAG pipeline using semantic chunking, hybrid search, vector embeddings, and ONNX-optimized inference, improving retrieval accuracy by 25%.
- Developed REST APIs and automated document ingestion workflows with vector-store indexing, data versioning, and LangSmith-based application tracing and evaluation.

AI Career Guidance Agent

OpenAI Assistants API, LangGraph, Streamlit, FAISS

- Developed an LLM-powered agent using function calling for resume parsing, skill-gap analysis, and personalized recommendations with FAISS semantic search and responsible-AI evaluation.